Matthew Ngoy

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Summary

CS senior focused on AI security and secure infrastructure; seeking a Summer 2027 internship.

Education

Kean University — Union, NJ

Bachelor of Science in Computer Science (Cybersecurity Option); GPA: 3.52 — *Expected May 2027*

- **Relevant Courses:** Ethical Hacking; Digital Forensics; Computer Security; Cryptography & Cryptanalysis; Data Structures; Analysis of Algorithms; Operating Systems; Database Management Systems.
- **Honors & Programs:** NSF IUSE Scholar (2025, 2026); CRA UR2PhD; CAHSI Travel Scholar; Dean's List; Upsilon Pi Epsilon; WELL Program; Young Envoys Scholarship.

Technical Skills

Languages: Python, Java, JavaScript, SQL, PHP, PowerShell
Security: OWASP LLM Top 10, MITRE ATLAS, OSINT, Nmap
AI & LLM: Ollama, OpenAI API (fine-tuning), LLM-as-judge evaluation, Claude Code, Codex
Tools & Frameworks: Linux (Ubuntu, Kali), Docker, Git, MySQL, SQLite, Node.js, Arduino/ESP8266
Coursework Exposure: TypeScript, Wireshark, Metasploit, Hashcat, Autopsy

Experience

Cybersecurity Research Intern — May 2026 – Present

Kean University Center for Cybersecurity — *Union, NJ / Wenzhou, China / Beijing, China*

- Investigated nonprofit cyber incidents through OSINT for the Non-Profit Cyber Incident Repository, classifying reports by CIA-triad impact, CISA sector, and TAG Cyber taxonomy.
- Design solar-power and rack-energy monitoring for Mobile ACDC, the Center's portable teaching data center.
- Built an Arduino/ESP8266 smart-home lab.
- Configured classroom computers for the Port Authority of NY/NJ STEaM Camp, preparing **30 laptops**.
- Taught cryptography, programming, and IP addressing across 12 lessons at STEaM Camp, reaching **60** high school students.

Peer Mentor / Code Samurai — Jan – May 2025; Sep 2026 – Present

Kean University Department of Computer Science (Code Samurai) — *Union, NJ*

- Guide students through Python, SQL, data structures, and debugging in one-on-one sessions; returned as a paid mentor after volunteering in Spring 2025.

Secretary — 2026

Women in CyberSecurity (WiCyS), Kean University Chapter — *Union, NJ*

- Coordinated guest-speaker scheduling and campus event planning for **PHD4FREE!**, developing the research and graduate-funding event from August through October 2026.
- Presented pathways into research, paid undergraduate programs, and graduate fellowships alongside Notre Dame PhD student Demetrius Hernandez, reaching **15 attendees**.

International Youth Delegate — Sep 2025

International Forum We, the Youth (IFWY) — *New York, NY / Seoul, Korea*

- Spearheaded project design and scope for SelectSolar, a proposal to divert solar panels from landfills and deliver low-cost, energy-efficient power solutions to educational and non-profit institutions.
- Selected for a fully funded fellowship at Cornell Tech, then voted 1 of 25 North American delegates for the IFWY Final Conference in Seoul, collaborating on international environmental policy and sustainable solutions.

Projects

KEAN CTF 2027 | *Python, Docker, CTFd, Cloudflare Tunnel, GitHub Actions* — Sep 2026 – Present

- Co-develop Docker-isolated CTF challenges and solve scripts with AI assistance and two teammates, including defenses intended to resist blind AI solvers.
- Automated challenge validation in GitHub Actions with AI assistance.
- Corrected an **unintended flag leak and broken hints** through an AI-assisted player audit.

LLM Red-Team Harness | *Python, SQLite, Ollama* — June 2026

- Built an AI-assisted pipeline for prompt-injection, jailbreak, and data-leak tests, saving model-judge verdicts and transcripts in SQLite and JSONL.

Research & Presentations

Independent Study, Red-Teaming HadAgent | *Advisor: Dr. Boyang Li* — Sep 2026 – Present

- Investigated verification guarantees by directing and interpreting AI-designed, AI-run tests of a copied HadAgent prototype; identified local **score-validation** and model-execution gaps.

Debate Coach LLM Evaluation | *CRA UR2PhD; JavaScript, Node.js, OpenAI API* — Feb – Apr 2026

- Helped fine-tune and evaluate LLM debate coaches with two teammates; contributed citation checks, participant interviews, and presentation, using an AI-assisted audit to examine source reliability.